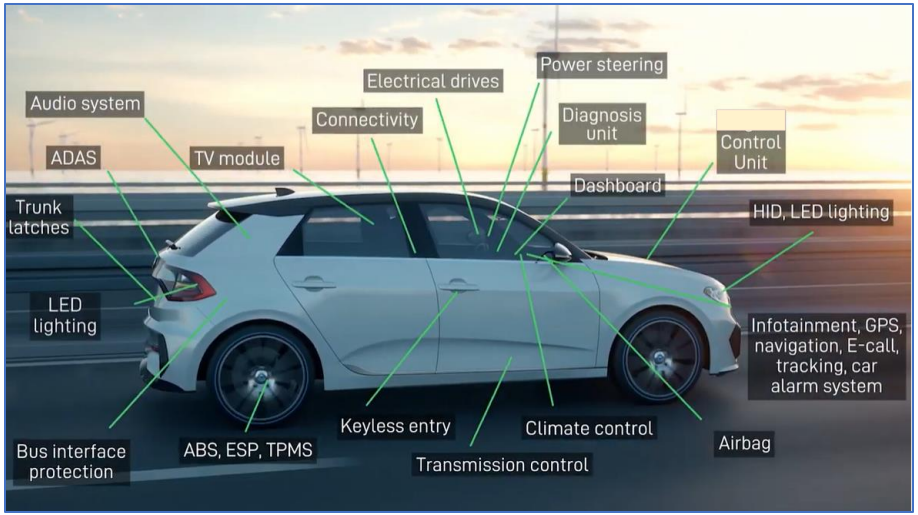


In Vehicle Network (IVN)

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Automotive Embedded Systems

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- There are multiple electronic control units on a given vehicle
- These controllers are carrying out independent functions, while exchanging information with each other on in-vehicle networks.
- Electronic control units for different systems/services have different requirements and require different technologies

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Control Systems Key points

Input conditions

Environmental conditions

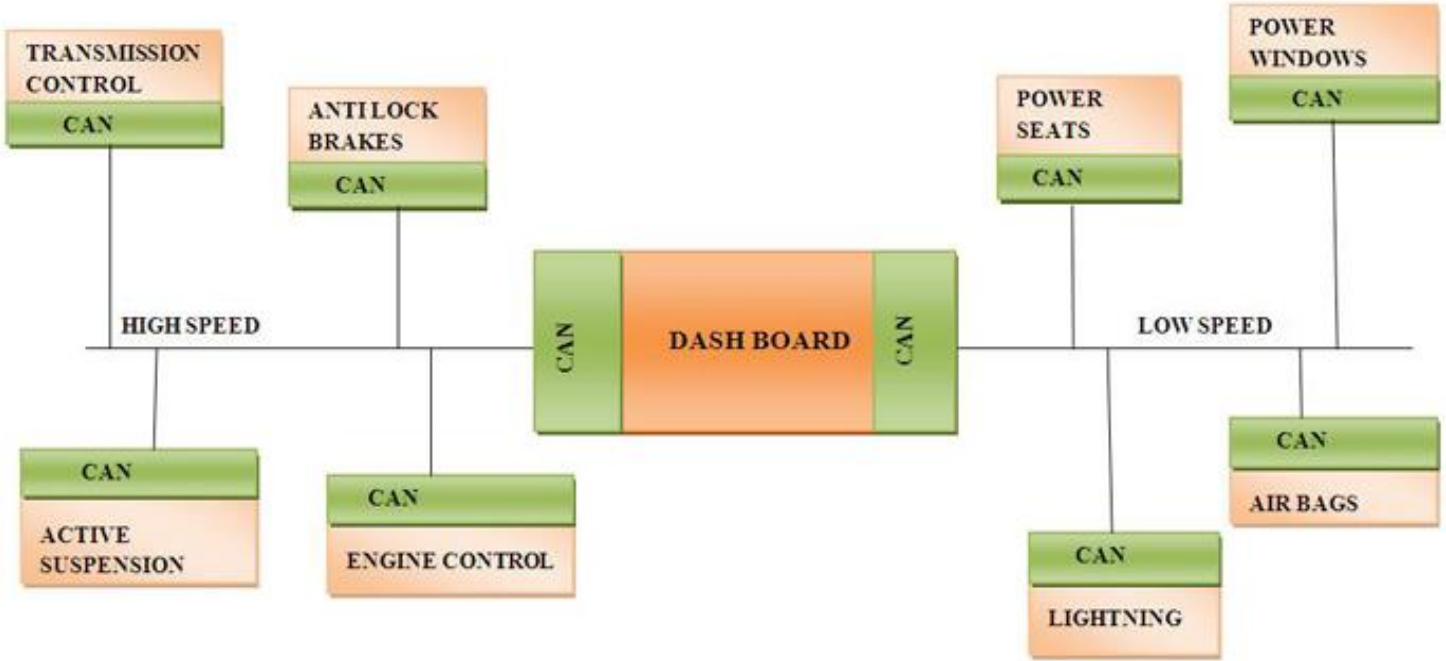
Output accuracy

Error detection

System state indication

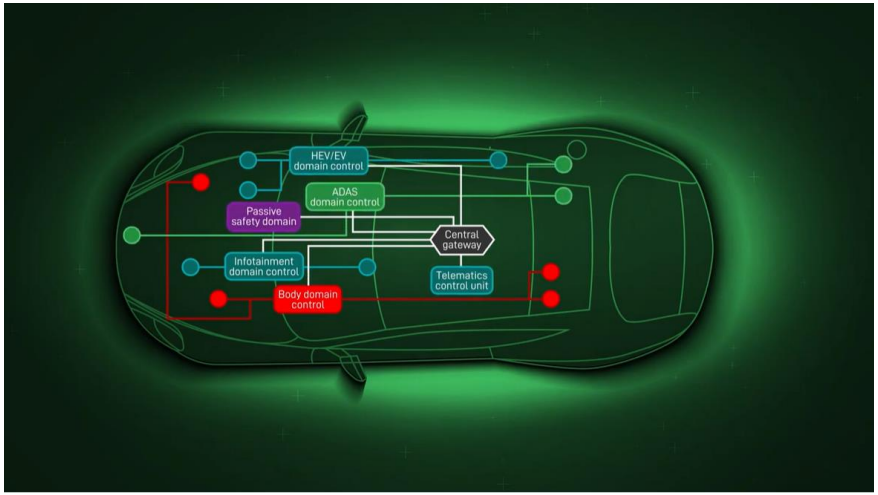
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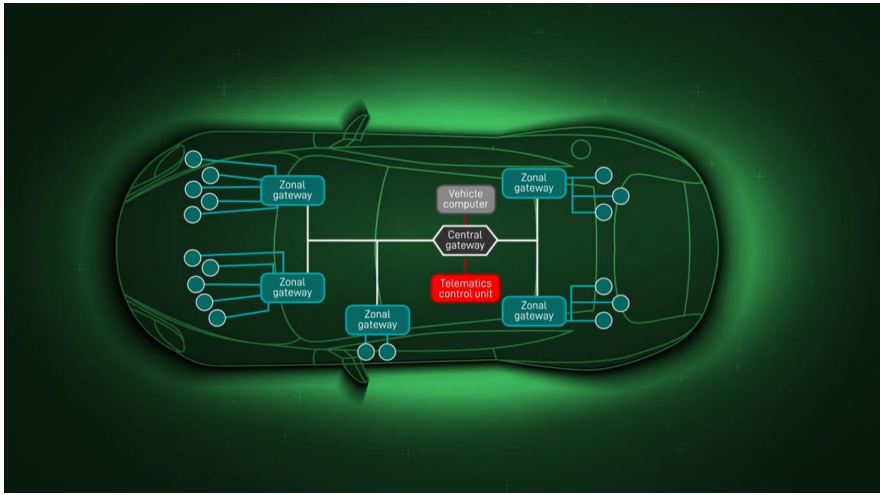
The Traditional Architecture



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Distributed Architectures

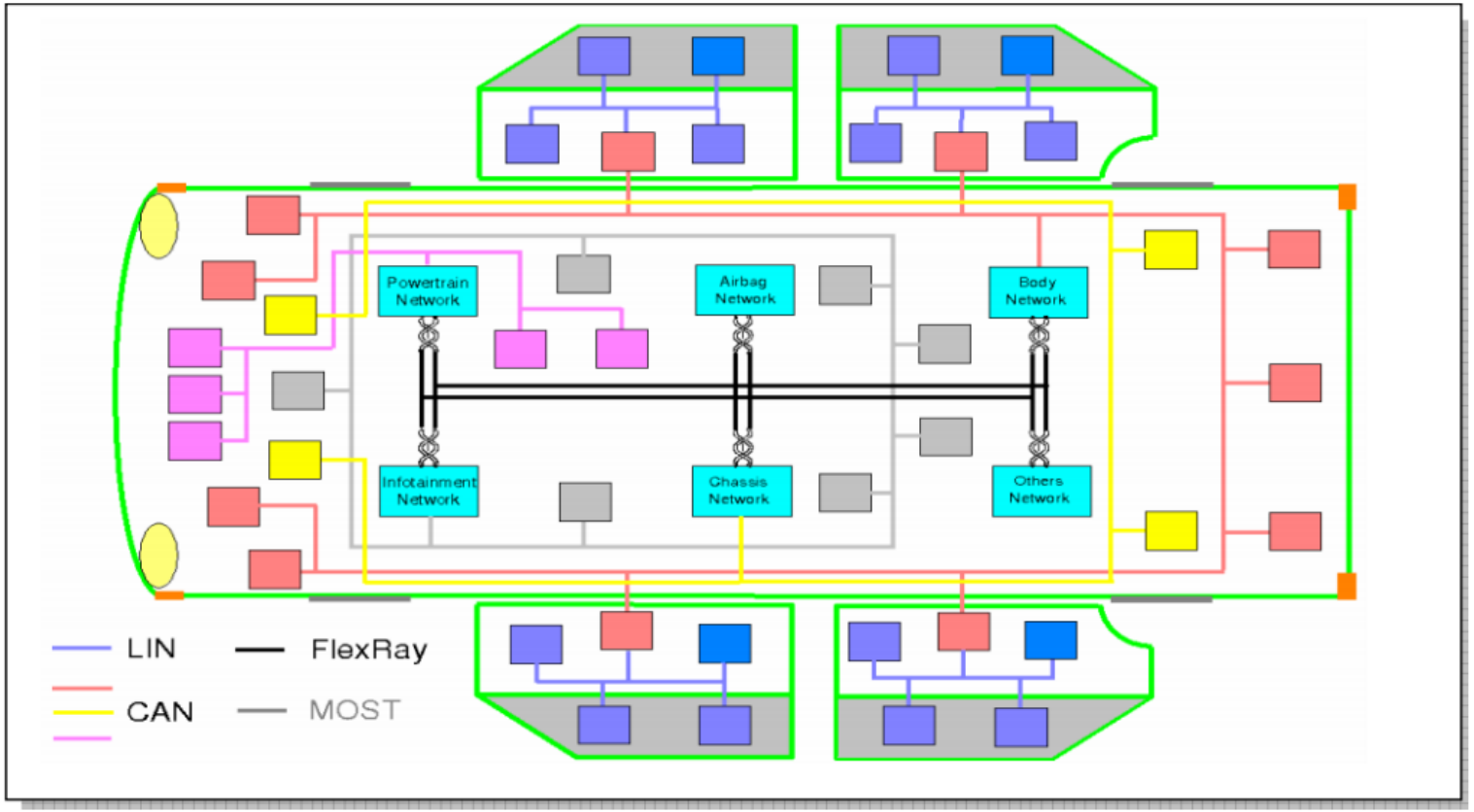




Distributed → Centralized

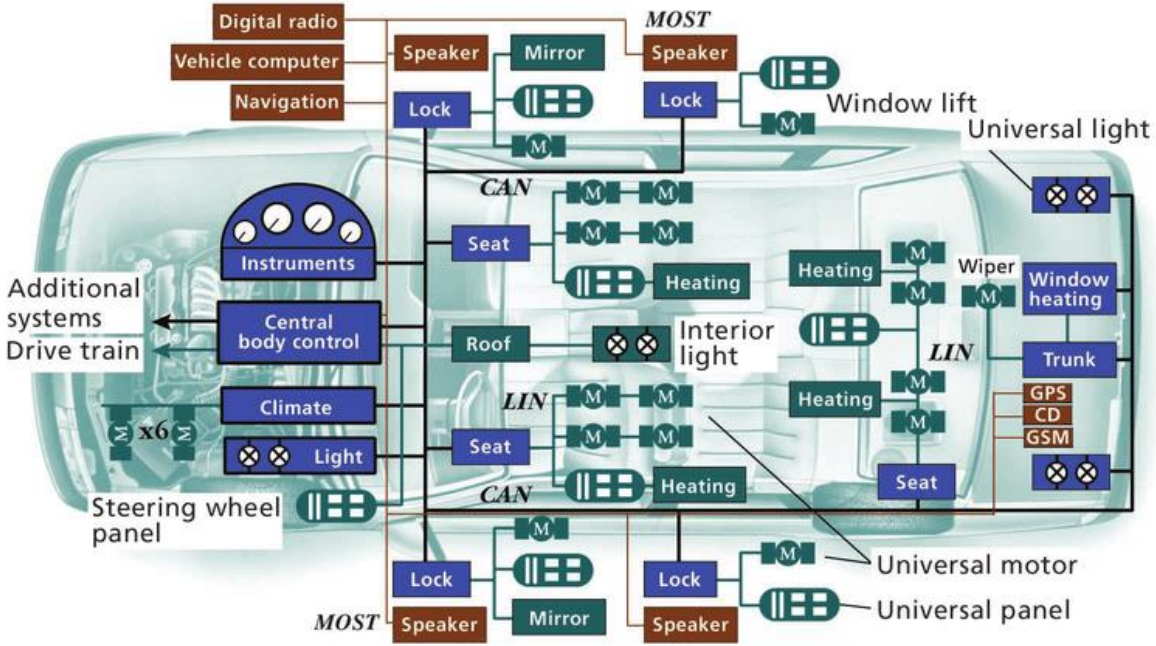
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Distributed Architectures



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Distributed Architectures



CAN Controller area network
GPS Global Positioning System
GSM Global System for Mobile Communications
LIN Local interconnect network
MOST Media-oriented systems transport

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IVN Protocols Comparison

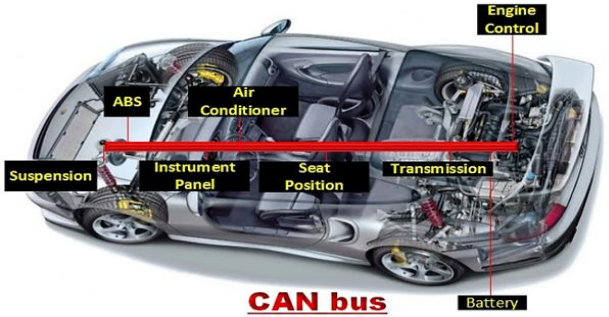
BUS	LIN	CAN	FLEXRAY	MOST
Cost/Node [$]	1.50	3.00	6.00	4.00
Used in	Subnets	Soft real-time	Hard real-time	Multimedia
Applications	Body	Chassis, Powertrain	Chassis, Powertrain	Multimedia, Telematics
Message transmission	Synchronous	Asynchronous	Synchronous & Asynchronous	Asynchronous & Synchronous
Data rate	20 kbps	1 Mbps	10 Mbps	24 Mbps
Physical layer	Single Wire	Dual-Wire	Dual-Wire (Optical-Fiber)	Optical-Fiber (Dual-Wire)
Latency jitter	Constant	Load dependent	Constant	Data stream
Extensibility	High	High	Low	High

Protocol	Bit-rate	Application	Domain	Standard
High CAN	125Kbps to 1Mbps	Real time critical applications e.g engine and braking systems	Powertrain and Chassis train	ISO 11898
CAN low	5kbps to 125kbps	Non-critical such as doors and windows	Body Domain	ISO 11898
CAN FD	Up to 10Mbps	Critical real time applications	Powertrain and Chassis train	ISO 11898
LIN	1kbps to 20 kbps	Non-critical applications	Body Domain	ISO 17987
FlexRay	up to 10 Mbps	Critical applications	Powertrain and Chassis	ISO 17458

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What is CAN ?

- **Controller Area Network** (CAN) was initially created by German automotive system supplier Robert Bosch in the 1983 for automotive applications.
- CAN became ISO standard 11898 and SAE J1939.
- CAN used especially for reliable data exchange between Electronic Control Units (ECUs) in the automobile.
- CAN controller chip were introduced by Intel and after that Philips.



Source: https://www.allaboutcircuits.com/technical-articles/introduction-to-can-controller-area-network

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CAN Features

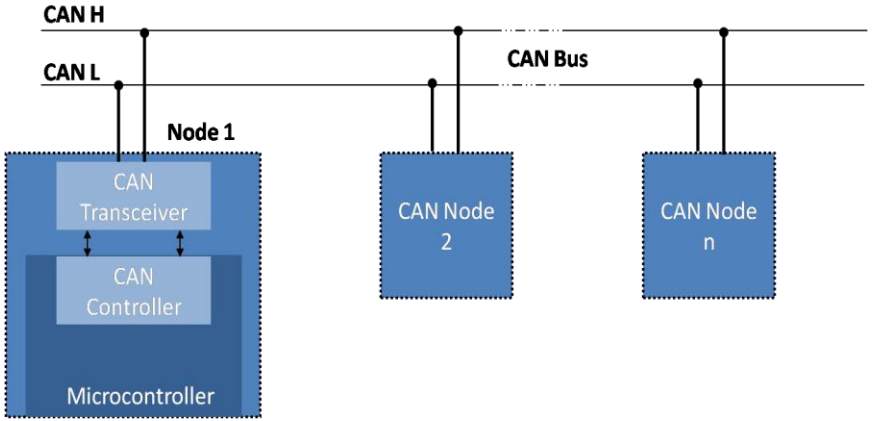
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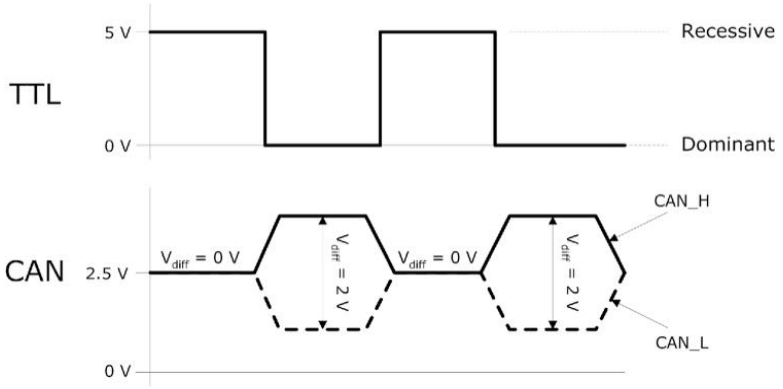
- CAN is half-duplex high speed serial communication protocol.
- CAN is a multi-master, multicast protocol.
- CAN is fast, robust communication.
- **The priority over bus is decided by the message not the node**.
- It is Message based, not address based
  - Messages are NOT transmitted from one node to another node based on addresses
  - New nodes can be added to a system without having to update all of the nodes with addressing information

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Why CAN Signal used ?

- CAN needs 2 lines, CAN_L, CAN_H.
- It is a differential line, where the signal is represented by a differences between CAN_L and CAN_H. If the difference is positive and larger than a certain minimum voltage it is called ‘Dominant’ and if the difference is zero it is called ‘Recessive.’

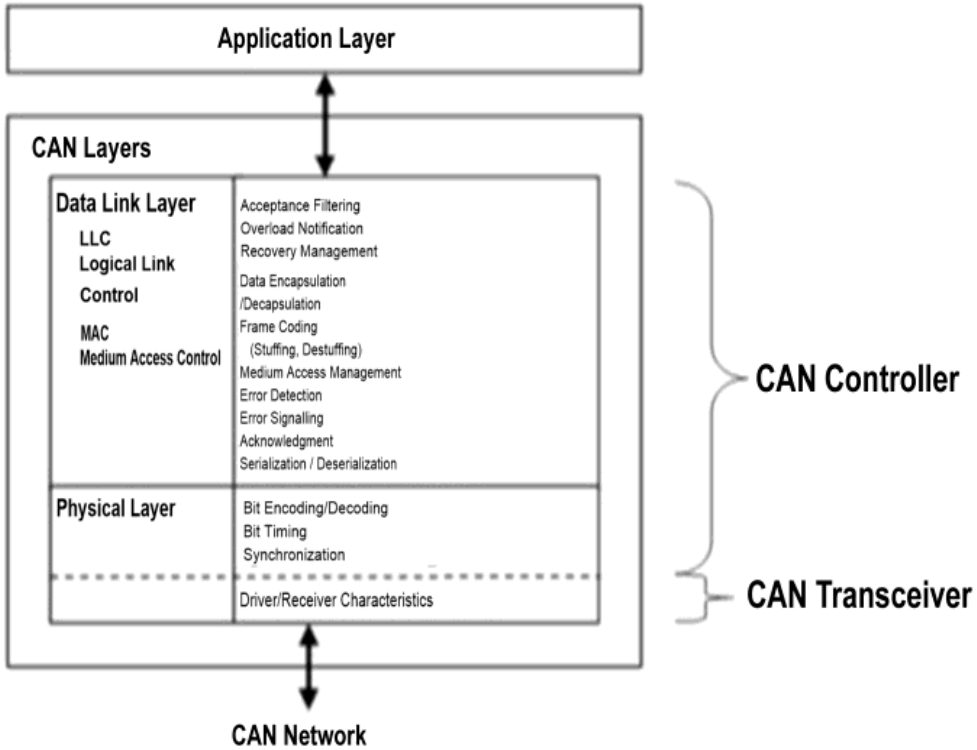




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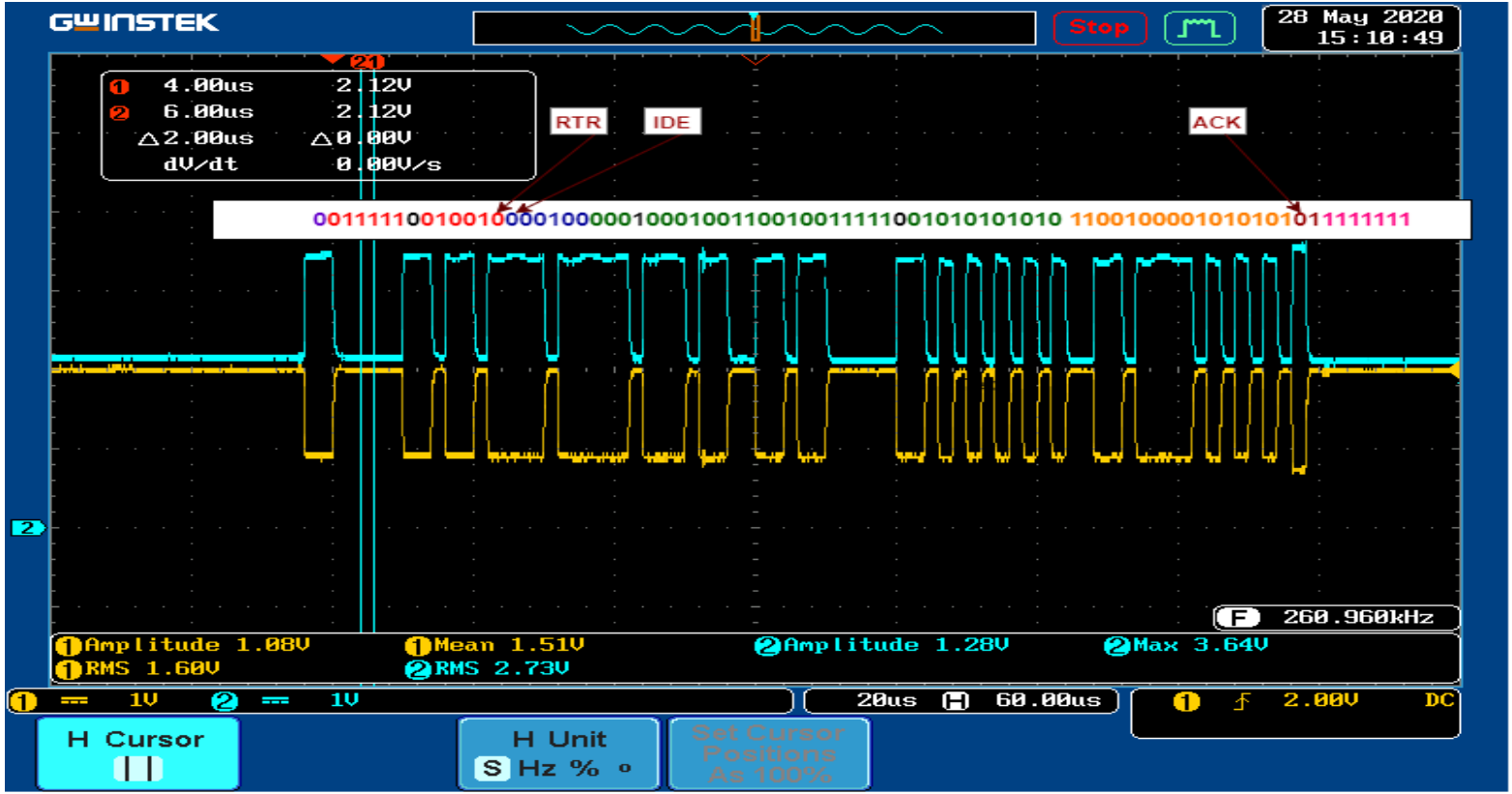
Data Link layer

- CAN Controller
  - Logical Link Control Sublayer (LLC)
    - Acceptance Filtering
    - Overload Notification
    - Recovery Management
- CAN Controller
  - Medium Access Control Sublayer (MAC)
    - Data Encapsulation/Decapsulation
    - Frame Coding(Stuffing, Destuffing)
    - Medium Access Management
    - Error Detection
    - Error Signaling
    - Acknowledgement
    - Serialization./Deserialization



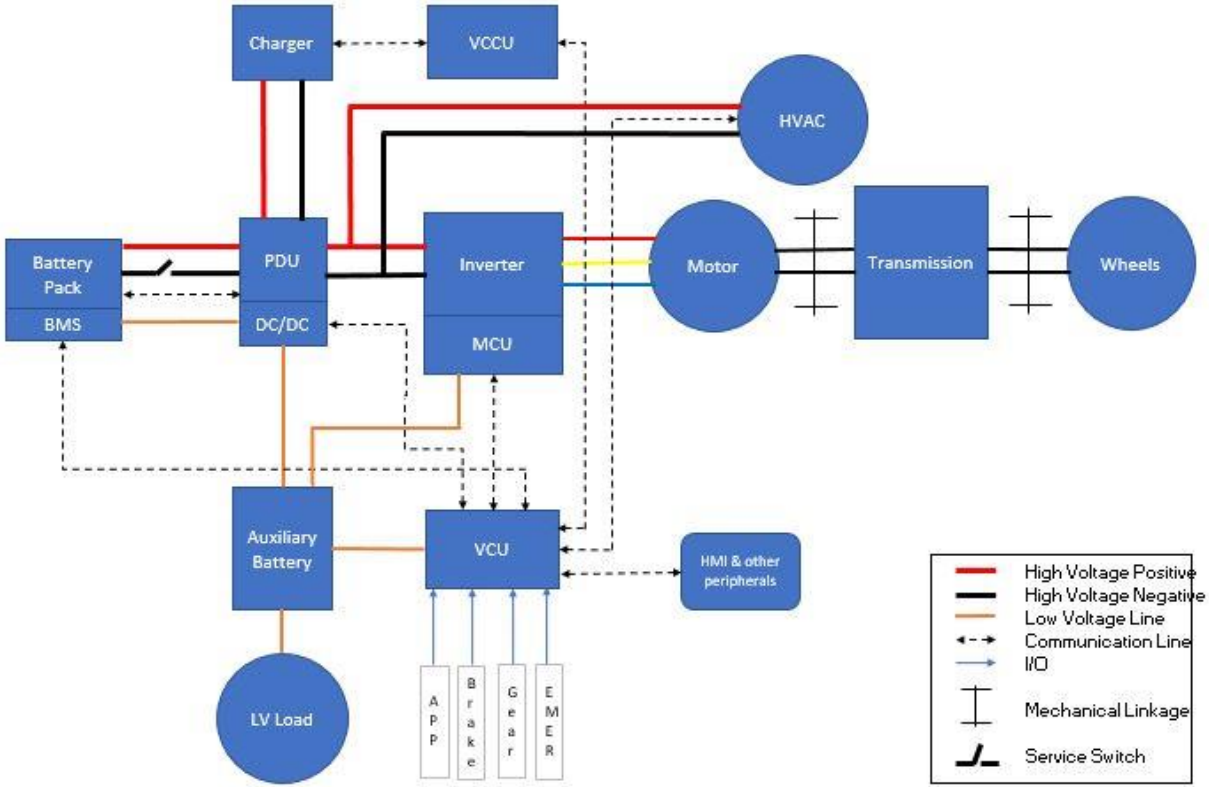
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Standard Frame Format



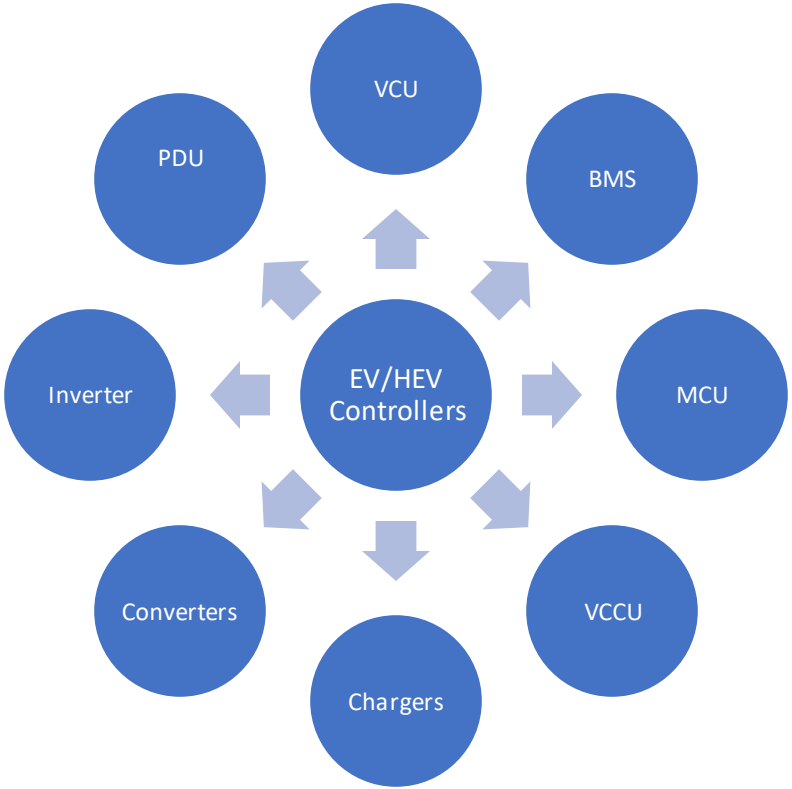
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EV Architecture – IVN



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EV/HEV Controllers



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Algorithms

- Driving Modes
  - ECO Mode
  - Sports Mode
- Battery Leakage Monitor
- Power Flow Monitor
- Plausibility Checks
- State Monitors
- Emergency Conditions
- Charging and Discharging Monitor
- Effective Regeneration

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System Level Functions

Should be Motor Controller ?



- Rotor Off-set Angle Compensation
- Input Voltage Monitoring
- Output Voltage Estimation
- Input Current Estimation
- Output Current Monitoring
- IGBT Temperature Monitoring
- IGBT Temperature Estimation
- Motor temperature Monitoring
- Inverter Input Power Estimation
- Motor Position & Speed Estimation
- Torque Capability Estimation
- Torque Management
  - Torque Control,
  - Torque Verification,
  - Active Damping,
  - Actual Torque Estimation
- HVIL
- Hill Hold
- LV Power Supply Management

- ePWT Cooling Management
- Electric Vacuum Pump Function
- LV Supply Management
- DC-DC Converter Control
- Charging Protocol Coordination Function
- HV Energy Management
- Regen Braking Function
- Vehicle Mode Management
- Accelerator Pedal Function
- Brake Pedal Function
- Torque Management
- Creep Torque Function
- Range Prediction
- ePWT supervisory Control
- HV Battery Cooling Function
- Cruise Control
- Functional Safety

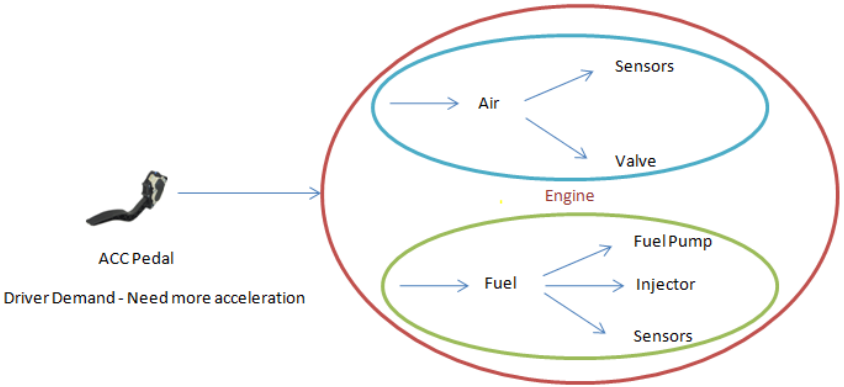
Voltage Monitoring
Cell Voltage Monitoring
Cell Balancing
Current Monitoring
Temperature Monitoring
SOC / SOH
Insulation Monitoring
HVIL
Power Supply Management (LV)
Fault Management
State Management
AC/DC Charging
Regeneration

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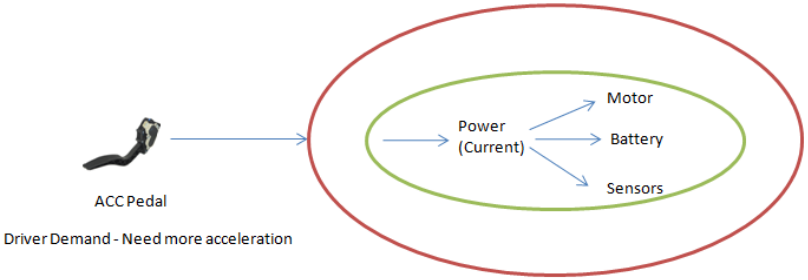
Driver Demands - Acceleration

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Conventional Vehicle



Electric Vehicle

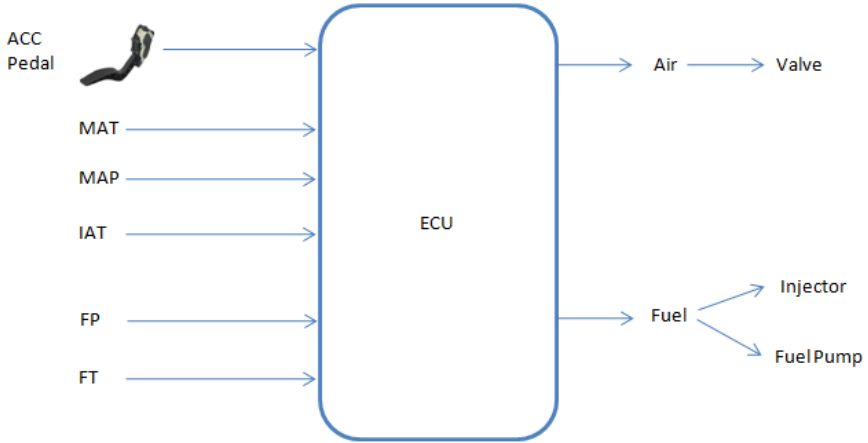


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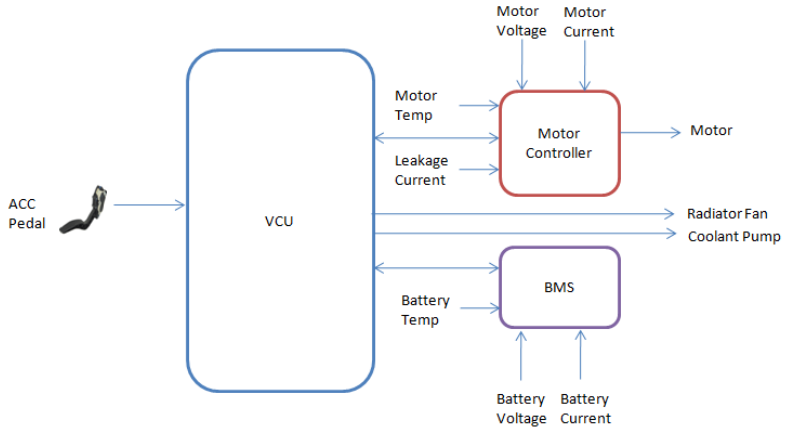
Driver Demands - Acceleration

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Conventional Vehicle



Electric Vehicle

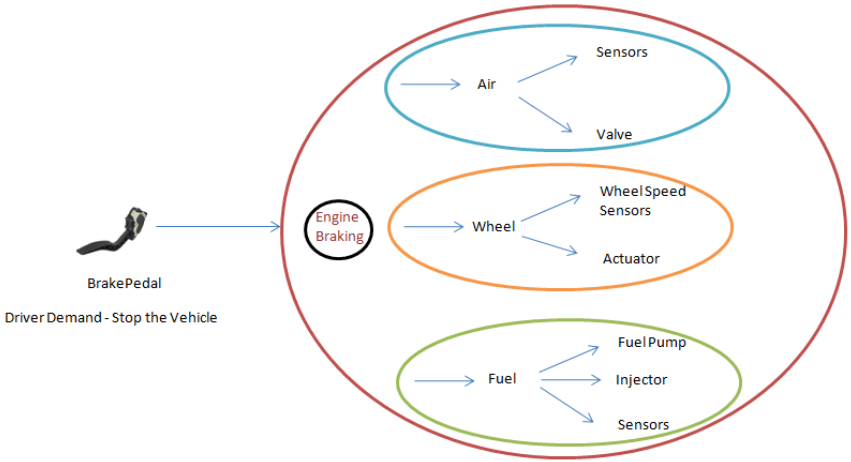


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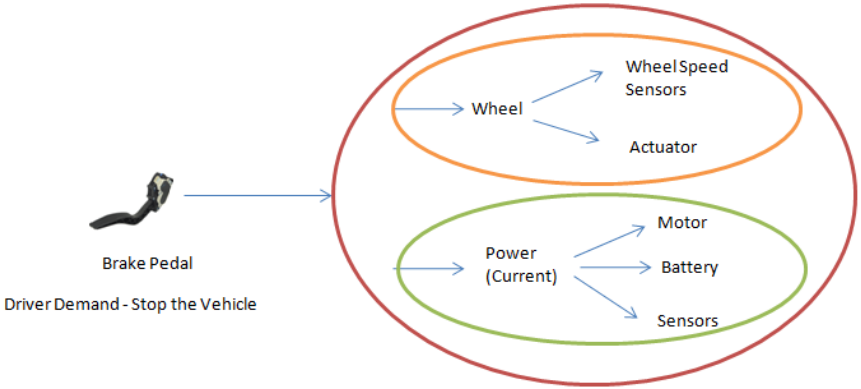
Driver Demands - Braking

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Conventional Vehicle



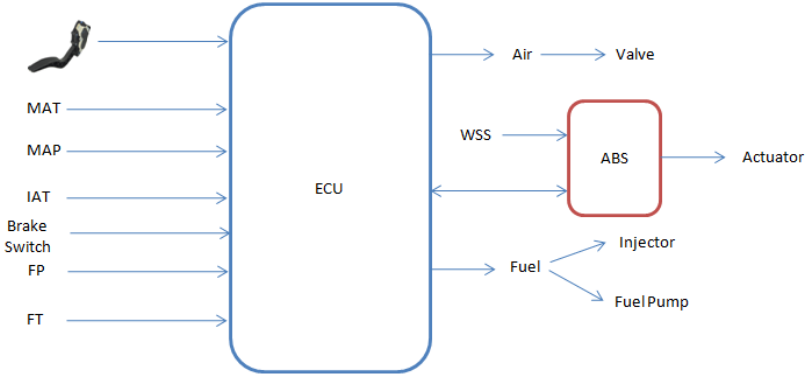
Electric Vehicle



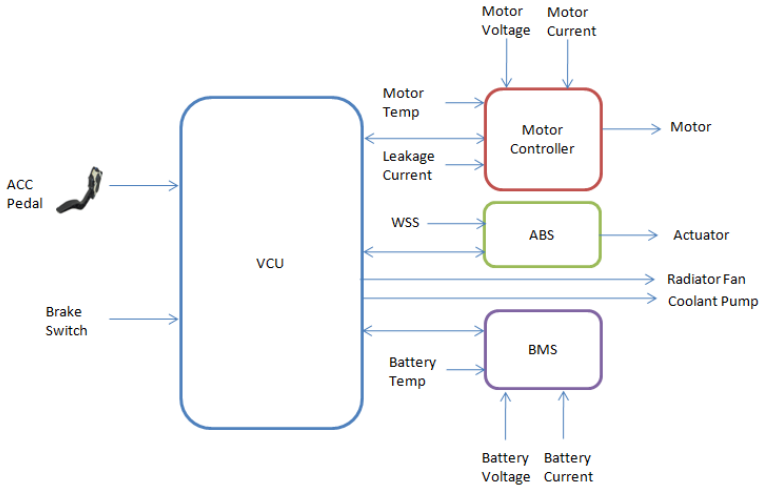
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Driver Demands - Braking

Conventional Vehicle

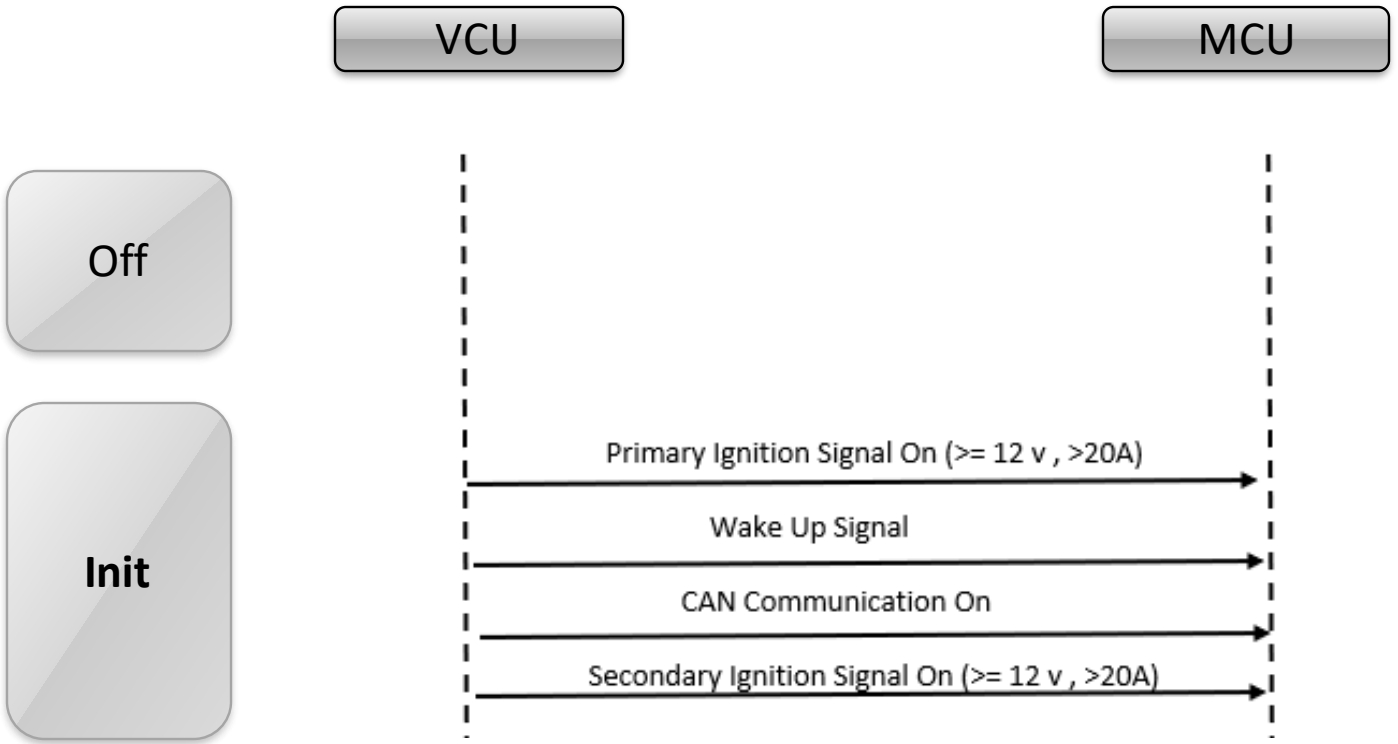


Electric Vehicle



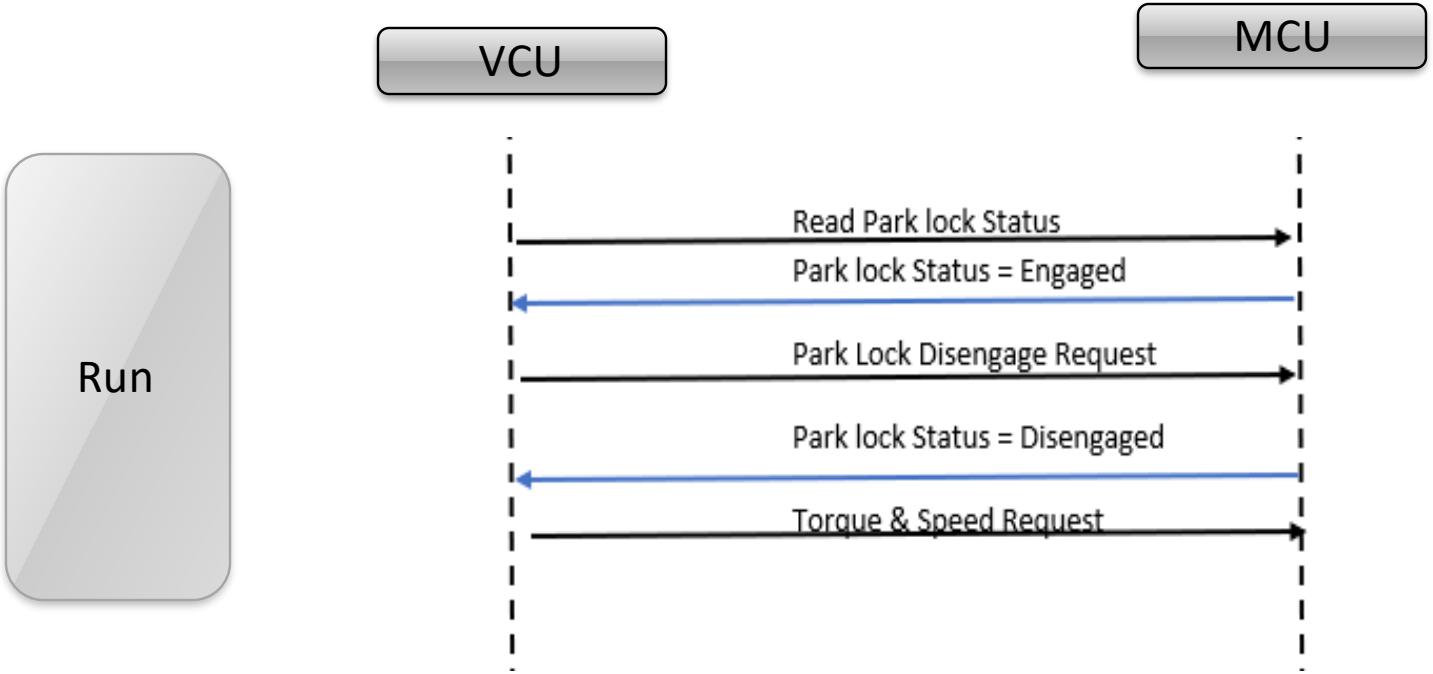
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VCU and MCU Handshaking



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VCU and MCU Handshaking



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Elements of ECU Interaction

Element	Typical Usage	Tools	Data Formats
Measurement	Typically done during testing and monitoring	Most OEMs use proprietary tools that can look at all internal engineering software parameters. This is made available to service engineers if basic troubleshooting does not solve the problem e.g.: INCA software	Parameter Lists – A2L Parameter Value - HEX Access Type – CCP or XCP
Calibration	Typically done during vehicle development and tuning	Most OEMs use proprietary tools to calibrate the settings and this is available only to field engineers	Same as above
Diagnostics	Typically done in after-market	Most OEMs use proprietary tools	Parameter List – ODX Value Decoding – ODX Access Type - UDS
Legislated Diagnostics	"OBD" – Mandatory to support standard tools		

Latency requirement of IVN

- From the driving safety distance: –55MPH/50FT: 0.613s; 110MPH/50FT: 0.306s
- From Human reaction time: 250ms
- 10us was mentioned in paper: https://ieeexplore.ieee.org/document/8315204, but it does not define it. It should be E2E latency for network perspective
- Use the fastest FlexRay as example :

–For minimum FlexRay pack size (9bytes), the serialization latency is 9*8/10M = 7.2us;
–For max Flexraypack size (262bytes), the serialization latency is 262*8/10M = 209us

- Also use FlexRay'scycle to predict, For real-time perspective, if there is any sporadic msg needs to be sent over FlexRay, the max delay by FlexRay is one cycle (about 1ms)
- On paper: https://ieeexplore.ieee.org/document/5981915

→ So, 1ms E2E delay for any p2p IVN communication is good enough!

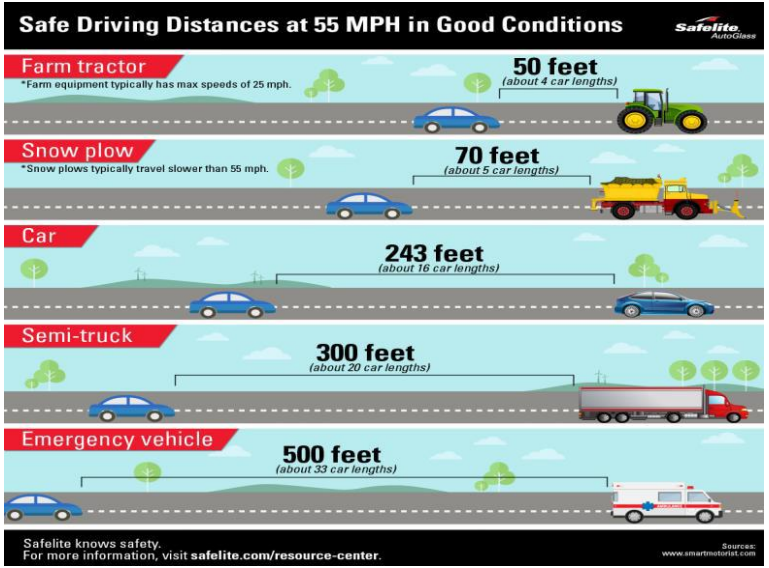


Table 2: Application constraints [9]

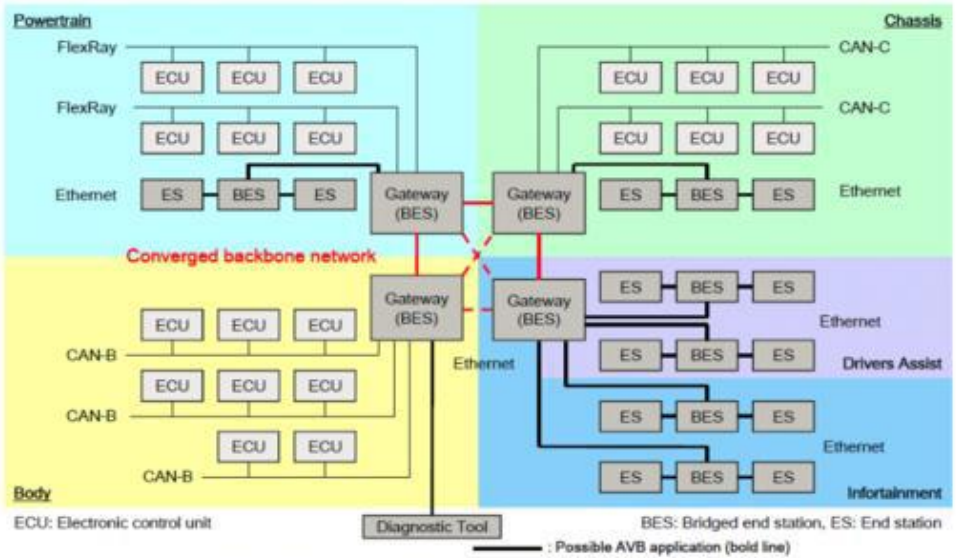
Priority Value	Traffic Type	Max. End-to-End Delay [ms]
3	Control	≤ 10 [16, 5]
2	Driver Assistance CAM	≤ 45 [5, 17]
1	Navigation	≤ 100
0	Multimedia	≤ 150 [18]

Industry proposal TSN

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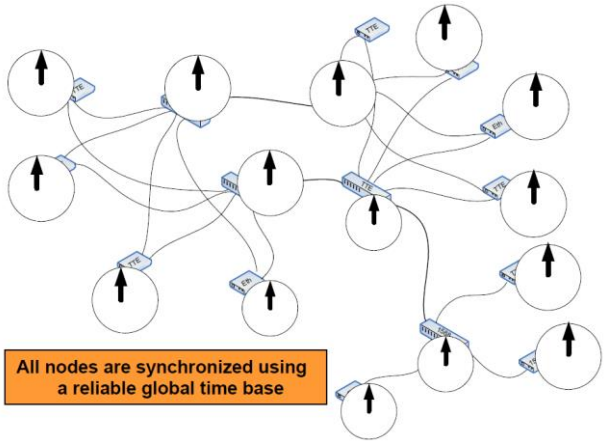
Use Case: Vehicular Network

- An example converged backbone network for the domain architecture

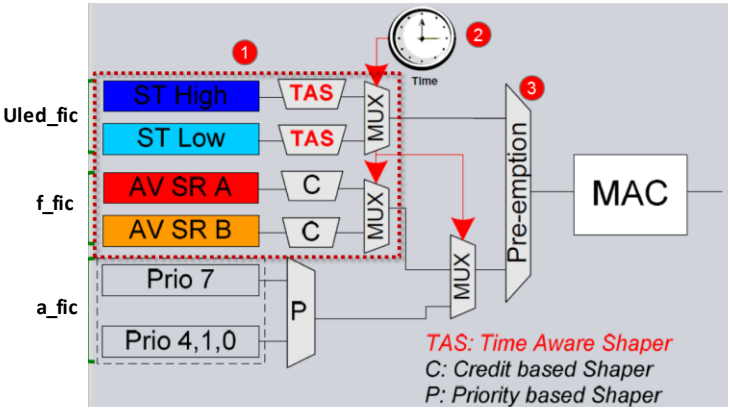


One possible application example of a future vehicular network

Source: IEEE 802.1 AVB TG presentation



All nodes are synchronized using a reliable global time base



Types of IP services for IVN

Service Type	Bandwidth	Latency	Jitter	Packet Loss	Use case
On-time (for Scheduled Traffic)	CIR Provided by APP is guaranteed by network	Most precise, Network guarantees E2E bounded latency	Approximately zero	Approximately Zero •m-path to prevent drop due to physical failure •Congestion-free •Lossless (due to queuing)	Synchronous communication: Critical sensor and control data
In-time (for Real-time Traffic)	CIR Provided by APP is guaranteed by network	Minimized, Network guarantees E2E bounded latency	½ of E2E bounded latency	Minimized •Congestion-free •Lossless (due to queuing) •Only drop due to physical failure	Asynchronous communication: Critical sensor and control data
Bandwidth Guaranteed (Bandwidth sensitive, but not time critical)	CIR Provided by APP is guaranteed by network Could reach PIR	Less important	Less important	Don't care	Un-critical data
Best Effort	Don't care	Don't care	Don't care	Don't care	Other data

Scheduled Traffic: Data size, data start time, data rate are fixed, used for critical sensor data and control data in polling mechanism
Real-Time Traffic: Data size and rate are fixed, but when the data starts is unknown, used for urgent sensor data change that polling mechanism did not catch

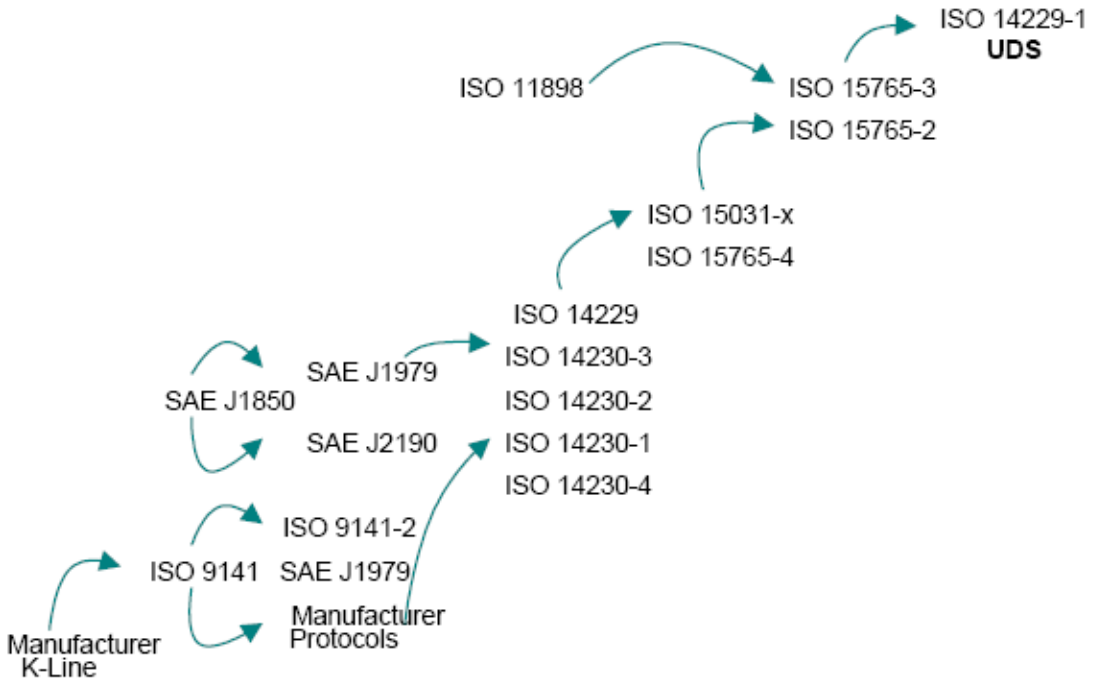
Communication Methodology

- Commands are sent via CMD
- Slave must respond with a RES or ERR
- Standard Commands
  - Setup Connection with Slave
  - Disconnect from Slave
  - Get Current Session Status from Slave
  - Get Communication Mode Info
  - Get Identification from Slave
  - Request to Save to Non Volatile Memory
  - Get Seed for Unlocking a Resource
  - Send Key for Unlocking a Resource
  - Set Memory Transfer Address in Slave
  - Upload From Slave to Master
  - Build Checksum over Memory Range

Data Acquistion Methods

- Polling
  - Send periodic requests
  - Command and Response Pair
- Data Acquisition
  - Configure a datalogger in the Slave with a sampling time
  - When sampling time is near, Slave will automatically collect and send data
- Stimulation
  - Periodic write of parameters to the Slave
  - Useful in calibration and tuning

ISO Protocol Evolution



UDS can be applied on physical layers – CAN, LIN, Ethernet, Flexray

ISO and SAE in OSI Framework

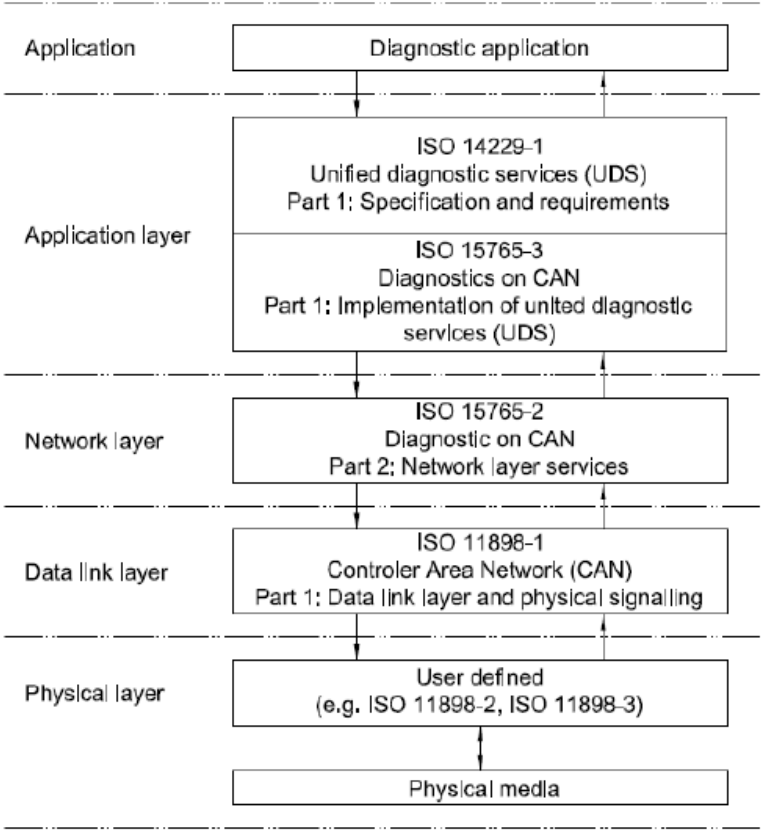
OSI Layer		MD & HD Standards & OBD Legislated	Pass Car & LD Standards	Pass Car & LD OBD Legislated
N/A	Diagnostic Connector	SAE J1939-13	ISO 15765-3 ISO 14229-1	ISO 15031-3
7	Application	SAE J1939-71/73 SAE J1939-81	ISO 15765-3	ISO 15031-5 (SAE J1979)
6	Presentation		User Defined (GMW3110, GGDS, KWP2000, etc.)	ISO 15031-5 (SAE J1979)
5	Session		ISO 15765-3	
4	Transport Protocol		ISO 15765-2	ISO 15765-2
3	Network Layer	SAE J1939-31	ISO 15765-2	ISO 15765-4
2	Data Link	SAE J1939-21 (ISO 11898-1)	ISO 11898-1	ISO 15765-4 (ISO 11898-1)
1	Physical Layer	SAE J1939-11/15	User Defined (J2284, J2411, ISO11898-2/3, etc.)	ISO 15765-4 (ISO 11898-2)

Passenger Car vs Medium/Heavy Duty

J1939 (MD & HD Truck)	Pass Car & LD Vehicles
All standards defined in J1939 parts	Many standards (primarily ISO14230, ISO14229 & ISO15031)
29 bit identifiers	Mostly 11 bit identifiers
Some messages broadcast regularly	No broadcast messages
Type of data in message defined within 29 bit identifier (PGN)	All information about data is in the data payload of the message
Many industry standard device names	Device names OEM dependent
Many parameters industry defined	Only emissions parameters standardized
Peer-to-Peer & broadcast requests	Same (physical / functional addressing)
4 byte fault codes	KWP = 2 byte, UDS = 3 byte fault codes
Four warning lamps defined	One warning lamp defined
Nine pin diag connector standard (J1939-13)	Sixteen pin diag connector standard (ISO15031-3 /J1962)

OSI view of UDS

- ISO14229-1 defines application layer requirements
- Any communication medium can be used at physical layer
- In automotive systems,
  - ISO 11898 is followed for physical layer characteristics
  - ISO 15765 is followed for Network layer and some of the application layer services



UDS Overview

- Application layer services are referred to as diagnostics services.
- These services are used to perform
  - Test
  - Inspection
  - Monitoring
  - Diagnostics of on-board vehicle servers
- The client is usually off-board tester, but can be an on-board tester as well in some systems.
  Example?
- More than one client can be present in an system

UDS Overview

- ISO 14229 has been established in order to define common requirements for diagnostic systems, whatever the serial data link is.
- The client -
  Usually referred to as external test equipment, uses the application layer services to request diagnostic functions to be performed in one or more servers.
- The Server -
  usually a function that is part of an ECU, uses the application layer services to send response data, provided by the requested diagnostic service, back to the client.

Diagnostics Terms

- Diagnostics data
  - Data that is located in the memory of an electronic control unit which may be inspected and/or possibly modified by the tester
- Diagnostics Routine
  - Routine that is embedded in an electronic control unit and that may be started by a server upon a request from the client
- Diagnostic service
  - Information exchange initiated by a client in order to require diagnostic information from a server or/and to modify its behavior for diagnostic purpose
- Server
  - Function that is part of an electronic control unit and that provides the diagnostic services
- ECU
  - Electronic control unit, containing at least one server

Diagnostics Services Info

- All of the services provided have same general structure
- Service primitives are used to exchange information
  - Service request primitive
  - Service request-confirmation primitive
  - Service indication primitive
  - service response primitive
  - service response-confirmation primitive
  - service confirmation primitive

Protocol Features

- The application layer protocol shall always be a confirmed message transmission
  - For each request from client, there shall be one or more corresponding responses from the server
  - Exception –
    - When functional addressing is used
    - Or request/indication specifies no response/confirmation shall be generated
    - For a few cases, negative response is not generated to not to burden system with unnecessary messages
- Positive response is generated by the server, if request from client satisfies all the requirements
- Negative response is generated if service request is not correct or is not supported by the server